MILITARY SPECIFICATION SHEET

# ELECTRON TUBE, RECEIVING

## TYPE 6005W 1/

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pentode, miniature, beam power amplifier

| | | |
|---|---|---|
| Outline | --- | 5-3 (EIA) |
| Base | --- | E7-1 |
| Envelope | --- | T5-1/2 |
| Cathode | --- | Coated unipotential |

Base connections:

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Pin No. | --- | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| Element | --- | g1 | k<br>g3 | h | h | a | g2 | g1 |

ABSOLUTE-MAXIMUM RATINGS:

| Parameter: | Ef | Eb | Ec1 | Ec2 | Ehk | Rgl | Ik | Ic | Pp | Pg2 | TE | Alt |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Unit: | V | Vdc | Vdc | Vdc | v | Meg | mAdc | mAdc | W | W | °C | ft |
| Maximum: | 6.9 | 275 | 0,-55 | 275 | 100 | 0.1 (see note 1) | 65 | 3.0 | 11.0 | 2.2 | 225 | (See note 2) |
| Minimum: | 5.7 | --- | --- | --- | --- | --- | --- | --- | --- | --- | --- | --- |
| TEST CONDITIONS: | 6.3 | 250 | -12.5 | 250 | 0 | --- | --- | --- | --- | --- | --- | --- |

GENERAL:

Qualification - Required

Reliable tube

1/ Formerly tube type 6005



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| METHOD | REQUIREMENT OR TEST | CONDITIONS | AQL (PERCENT DEFECTIVE) | INSPECTION LEVEL OR CODE | SYMBOL | LIMITS MIN | LIMITS MAX | UNIT |
|---|---|---|---|---|---|---|---|---|
| | Quality conformance inspection, part 1 | | | | | | | |
| 1301 | Heater current | | 0.4 | II | If | 420 | 480 | mA |
| 1336 | Heater-cathode leakage | | 0.4 | II | Ihk | --- | 20 | µAdc |
| 1266 | Total grid current | Rg1 = 0.5 Meg (see note 3) | 0.4 | II | Ic | 0 | -1.0 | µAdc |
| 1256 | Eletrode current (1) (anode) | See note 3 | 0.4 | II | Ib | 33 | 57 | mAdc |
| 1256 | Electrode current (screen) | | --- | --- | Ic2 | --- | --- | mAdc |
| 1256 | Electrode current (screen) | | 0.4 | II | Ic2 | 0 | 7.5 | mAdc |
| 1341 | Power output (1) | Esig = 8.8 Vac; RL = 5,000 ohms | 0.4 | II | Po | 3.6 | --- | W |
| 1201 | Short and discontinuity detection | | 0.4 | II | --- | --- | --- | --- |
| | Quality conformance inspection, part 2 | | | | | | | |
| 1256 | Electrode current (2) (anode) | Ec1 = -60 Vdc | 2.5 | I | Ib | --- | 200 | µAdc |
| 1211 | Insulation of electrodes | | 2.5 | I | R | 150 | --- | Meg |
| 1306 | Transconductance | | 2.5 | I | Sm | 3,000 | 5,200 | µmhos |
| 1341 | Power output (2) | Ef = 5.7 V; Esig = 8.8 Vac; RL = 5,000 ohms | 2.5 | I | ΔPo Ef | --- | 15 | % |
| 1266 | Grid emission | Ef = 7.5 V; Ec1 = -50 Vdc; Rg1 = 0.5 Meg; Rk = 250 ohms (see note 4) | 2.5 | I | Ic1 | 0 | 4.0 | µAdc |
| 2201 | Noise and microphonics | Ef = 6.3 Vac; Ecal = 140 mVac; Rp = 2,000 ohms; Rg1 = 0.1 Meg | 2.5 | I | --- | --- | --- | --- |
| 1266 | Primary grid emission | Eg2 = 135 Vac, 60 Hz; Eb = 0 (see note 5) | 6.5 | S3 | Ic2 | --- | 750 | µAdc |
| 1331 | Direct-interelectrode capacitance | No shield | 6.5 | Code E | Cg1p<br>Cin<br>Cout | ---<br>6.4<br>6.0 | 0.8<br>9.6<br>11.0 | pF<br>pF<br>pF |
| 1031 | High-frequency vibration | Ec1 = -25 Vdc, Rp = 2,000 ohms | 6.5 | Code E | Ep | --- | 300 | mVac |
| 1041 | Shock | 450 G; Ehk = 100 Vdc; Ec1 = 0; Rk = 250 ohms (see note 6) | 6.5 | See note 7 | --- | --- | --- | --- |
| 1031 | Vibration fatigue | | 6.5 | See note 7 | --- | --- | --- | --- |
| --- | Shock and vibration-fatigue test end points: | | | | | | | |
| 1031 | High-frequency vibration | | --- | --- | Ep | --- | 500 | mVac |
| 1336 | Heater-cathode leakage | | --- | --- | Ihk | --- | 50 | Adc |
| 1341 | Power output (1) | | --- | --- | Po | 3.0 | --- | W |
| 1266 | Total grid current | | --- | --- | Ic1 | 0 | -4.0 | µAdc |
| 1121 | Base strain | See note 8 | --- | --- | --- | --- | --- | --- |



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| METHOD | REQUIREMENT OR TEST | CONDITIONS | AQL (PERCENT DEFECTIVE) | INSPECTION LEVEL OR CODE | SYMBOL | LIMITS MIN | LIMITS MAX | UNIT |
|---|---|---|---|---|---|---|---|---|
| | Quality conformance inspection, part 2 -Continued | | | | | | | |
| 2126 | Envelope strain | | 2.5 | I | --- | --- | --- | --- |
| 1105 | Permanence of marking | | --- | --- | --- | --- | --- | --- |

| METHOD | REQUIREMENT OR TEST | CONDITIONS | SYMBOL | LIMITS MIN | LIMITS MAX | UNIT |
|---|---|---|---|---|---|---|
| | Quality conformance inspection, part 3 | | | | | |
| 1506 | Heater-cycling | Ef = 7.5 V; Ehk = 135 Vdc; Ec1 = Ec2 = Eb = 0 | --- | --- | --- | --- |
| --- | Heater-cycling life-test end point: | | | | | |
| 1336 | Heater-cathode leakage | | Ihk | --- | 40 | µAdc |
| 1516 | Stability life | Ec1 = 0; Ehk = 135 Vdc; Rg1 = 0.5 Meg; Rk = 250 ohms; TA = room | --- | --- | --- | --- |
| --- | Stability life-test end point: | | | | | |
| 1341 | Change in power output (1) of individual tubes | | ΔPo | --- | 10 | % |
| 1501 | Intermittent life | Ec1 = 0; Ehk = 135 Vdc; Rg1 = 0.5 Meg; Rk = 250 ohms; TA = room TE = 225°C (min) (see note 9) Group E | --- | --- | --- | --- |
| --- | Intermittent life-test end points (1,000 hours): | | | | | |
| --- | Inoperatives | | --- | --- | --- | --- |
| 1266 | Total grid current | | Ic1 | 0 | -1.0 | µAdc |
| 1301 | Heater current | | If | 420 | 490 | mA |
| 1341 | Change in power output (1) of individual tubes | | ΔPo t | --- | 25 | % |
| 1341 | Power output (2) | | ΔPo Ef | --- | 20 | % |
| 1336 | Heater-cathode leakage | | Ihk | --- | 20 | µAdc |
| 1211 | Insulation of electrodes | | R | 50 | --- | Meg |

NOTES:

1. This value is for operation under fixed bias conditions. With cathode bias, Rg1 may be 0.5-megohm maximum.
2. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage.
3. This test shall be performed at the conclusion of the holding period.
4. Prior to this test, tubes shall be preheated a minimum of 5 minutes with all sections operating at the conditions specified below. The 3-minute test shall not be permitted. Test at specified conditions within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid-emission test.

| Ef V | Ec1 Vdc | Ec2 Vdc | Eb Vdc | Rk Ohms | Rg1 Meg |
|---|---|---|---|---|---|
| 7.5 | 0 | 250 | 250 | 250 | 0.5 |

5. Operate the TUT for 5 minutes at test conditions prior to the primary screen-grid emission test. Adjust Ec1 to provide Ic2 = 13.4 mAdc on the positive half cycles.
6. A grid resistor of 0.1 megohm shall be added; however, this resistor shall not be used when a thyratron-type short indicator is employed.
7. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. When one lot has passed, the 12-month rule shall apply. MIL-STD-105, sample size code letter E, shall apply.

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NOTES: -Continued.

8. Acceptance sampling procedure shall be in accordance with "Base-strain test, miniature, sampling (method 1121)", except that data covered in "Acceptance and rejection criteria" shall be modified as follows:

   (a) Accepted if not more than one defective for class "A", "B", or "C" defects, respectively (see method 1121), or if not more than a total of two defectives are found in the sample.

   (b) Rejected if two or more defectives for class "A", "B", or "C" defects, respectively, or if a total of three or more defectives are found in the sample.

9. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a TUT having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

10. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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